

Teacher Guide

FireSat CubeSat Mission
STEMSat Educational CubeSat Program

1. Mission Overview

In this activity, students investigate how environmental data can be used to detect hazards such as wildfires and extreme heat. Students build and operate a CubeSat-inspired device equipped with a BME688 environmental sensor and an analog light sensor module. These sensors allow the CubeSat to measure temperature and light intensity.

Students conduct experiments such as heat tests, light exposure tests, and combined condition tests to explore how environmental changes can indicate hazardous conditions. Sensor data is transmitted through the Serial Monitor or HC-05 Bluetooth module, simulating how real satellites send telemetry data to a ground station.

Through testing and data analysis, students explore how patterns in environmental data can be used to detect hazards, while also applying the engineering design process to evaluate and improve their CubeSat system.

2. Target Grade Level

Grades: 6–8 (Middle School)

3. Time Requirements

Activity	Time
Mission introduction	10 minutes
Hardware assembly	15–20 minutes

Code upload and sensor verification	10 minutes
Environmental experiments	25 minutes
Data analysis and discussion	15 minutes
Engineering redesign discussion	10 minutes

Total Time: ~90 minutes

This activity can also be divided into two class periods.

4. Learning Objectives

By the end of the activity, students will be able to:

1. Explain how sensors measure environmental conditions
2. Investigate how temperature and light relate to environmental changes
3. Interpret sensor data to identify potential hazards
4. Understand how satellites monitor Earth for dangerous conditions
5. Apply engineering design principles to improve a system

5. NGSS Alignment

Engineering Design

Standard	Application
MS-ETS1-2	Students compare results from multiple environmental tests and evaluate system performance
MS-ETS1-3	Students analyze sensor data to identify patterns and design improvements
MS-ETS1-4	Students build and test a CubeSat model and refine their design through experimentation

Science Standards

Standard	Application
MS-ESS3-2	Students use environmental data to discuss hazard monitoring
MS-ESS3-3	Students design a method to monitor environmental conditions
MS-ESS3-5	Students analyze environmental data to identify patterns and changes

6. Materials

Per Student Group (2–4 students)

- Arduino Uno
- BME688 environmental sensor
- Light sensor module (ALS-PT19)
- HC-05 Bluetooth module
- Female-to-female jumper wires
- Cable holders (optional)
- USB cable
- Computer with Arduino IDE
- CubeSat frame

Instructor Materials

- Pre-installed Arduino libraries
 - Pre-tested sensors
 - Printed mission guides for each student
 - Flashlights or adjustable light sources
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7. Pre-Class Setup

Install Arduino IDE

Install on all computers.

Download:

<https://www.arduino.cc/en/software>

Install Required Libraries

FireSat requires the following libraries:

- Adafruit_BME680
- Adafruit_Sensor

How to install/check:

1. Open Arduino IDE
 2. Go to Tools → Manage Libraries
 3. Search: Adafruit BME680
 4. Install if not already installed
 5. Search: Adafruit_Sensor
 6. Install if not already installed
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Test Components / Make Example Model

Before class:

1. (Optional but highly recommended) Pre-attach wiring to sensors (refer to wiring instructions in the mission guide) and optionally use a 3D printed connector holder to hold wires in place and keep students from mixing up wiring for sensors when connecting to the Arduino.
 2. Follow the wiring instructions in the Mission Guide to attach sensors and upload code to one Arduino to ensure everything is working/to create an example model for students
 3. Verify readings appear in the serial monitor
 4. Verify that: Temperature readings change when the sensor is warmed, and light readings change when light conditions change
 5. Pre-check that the Bluetooth module can connect and print out values wirelessly to the serial monitor
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Prepare Test Stations

Set up areas for experiments such as:

- Flashlight testing station
 - Low-light or shaded area
 - Safe area for heating sensor (e.g., using hands)
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8. Running the Activity

Step 1: Introduce the Mission

Ask students:

- How do satellites detect wildfires or extreme heat?
- Why is it important to monitor environmental conditions?
- What types of data might indicate a hazard?

Explain that students will simulate environmental monitoring using their CubeSat system.

Step 2: Assemble the CubeSat

- Students connect the BME688 sensor and light sensor module to the Arduino.
 - Instructor checks wiring before power is applied.
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Step 3: Upload the Code

Students upload the mission code and open the Serial Monitor.

Confirm that values change when:

- The sensor is warmed
 - Light levels increase or decrease
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9. Environmental Experiments

Students perform three environmental tests.

Test	Purpose
Heat Test	Simulate rising temperature
Light Test	Simulate bright environmental conditions
Combined Test	Simulate potential hazard conditions

Students record sensor readings during each test.

10. Expected Results

Typical observations:

Sensor	Observation
BME688	Temperature increases when warmed
Light Sensor	Values increase with brighter light

Students should observe that higher temperature and light levels may indicate potential hazard conditions.

11. Data Analysis

Students analyze:

- Which test produced the highest temperature
- Which test produced the highest light level
- Which conditions triggered warning or hazard states

Instructor discussion prompts:

- Why might high temperature indicate a hazard?
 - Why is combining multiple sensor readings useful?
 - How could satellites use this data to detect wildfires?
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12. Engineering Design Discussion

Students evaluate the CubeSat design.

Possible design improvements:

- Add additional sensors
- Improve sensor placement

- Increase sensitivity or detection thresholds
- Enhance data interpretation methods

This reinforces the engineering design cycle.

13. Troubleshooting Guide

Problem: Temperature readings do not change

Possible cause:

- Incorrect wiring or poor connection

Fix:

- Verify BME688 connections (SDA/SCL, power, ground)

Problem: Light readings do not change

Possible cause:

- Incorrect analog wiring

Fix:

- Verify A0 connections

Problem: No data output

Possible cause:

- Serial Monitor or Bluetooth not connected

Fix:

- Check COM port and connections
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14. Safety Considerations

- Do not use open flames or hot objects
 - Use only safe methods (hands) to warm sensors
 - Avoid shining bright lights directly into eyes
 - Keep wires secured during experiments
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15. Extension Activities

Extension 1: Hazard Threshold Design

Students define their own thresholds for NORMAL, WARNING, and HAZARD

Extension 2: Multi-Sensor Analysis

Students compare temperature vs light contributions to hazard detection

Extension 3: Environmental Mapping

Students collect data in different locations and compare results

16. Real-World Space Connection

Satellites monitor Earth to detect environmental hazards such as wildfires and extreme heat. Engineers use data from temperature and light sensors to identify dangerous conditions and provide early warnings.

Similar systems are used in:

- Wildfire detection satellites
- Climate monitoring systems
- Earth observation missions

The FireSat mission allows students to simulate how satellites detect and respond to environmental hazards.

17. Mission Success Criteria

The mission is successful if:

- CubeSat sensors detect changes in temperature and light
- Students perform multiple environmental tests
- Students interpret environmental data
- Students propose improvements to the CubeSat design